2ND SEMESTER LAB REPORT

TITLE : TO VERIFY KIRCHHOFF'S VOLTAGE LAW & CURRENTS LAW DIFFERENTCONFIGURATION OF DC CIRCUITS



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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 5

Lab Title: VERIFCATION OF OHM'S LAW

Objective: 1TO VERIFY KIRCHHOFF'S VOLTAGE LAW & CURRENTS LAW IN DIFFERENT Circuits

CONFIGURATION OF DC CIRCUITS -

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 18/March/2025 Signature:

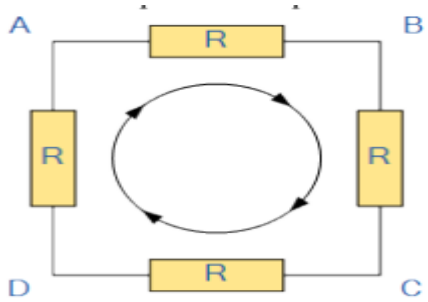
LABORATORY
EXPERIMENT
NO. 5

TO VERIFY KIRCHHOFF'S VOLTAGE LAW & CURRENTS LAW IN DIFFERENT CONFIGURATION OF DC CIRCUITS

OBJECTIVE:

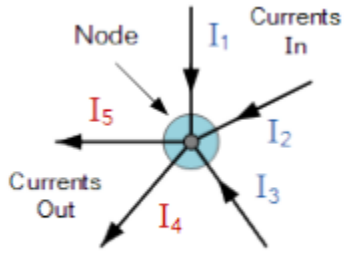
1. To verify Kirchhoff's voltage law for dc circuits.
2. To verify Kirchhoff's Current law for dc circuits.

1. Kirchhoffs Voltage Law (KVL)



$$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$$

2. Kirchhoffs Current Law (KCL)

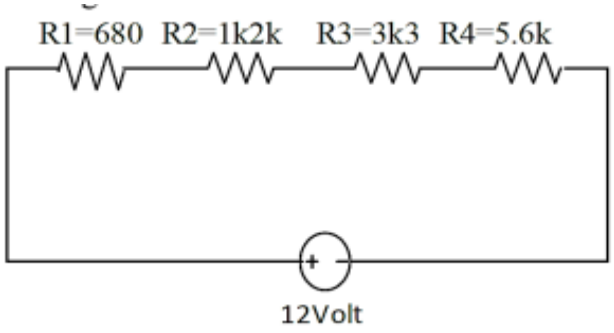


$$I_1 + I_2 + I_3 + (-I_4 + -I_5) = 0$$

LAB TASKS:

TASK 1. Solve the following circuit by KVL:

(a) Connect the following circuit on breadboard.



R1=6.8K Ω



Four resistors of given values are connected in series circuit on breadboard

b) Before connecting the power supply measure the total resistance of the circuit.

Req(measured) =15.9K Ω

Req(calculated) =680 Ω + 1.2 kΩ + 3 kΩ + 5.6 kΩ =10.480K Ω

This is measured equivalent resistance of circuit ion four resistors are connected in series
Measured Resistance


c) Using the measured value of the resistance calculate the current I, voltage across R1, R2, R3 and R4. Record all these values along with the source voltage in the calculated section of the following table. I (calculated) = 4.92mA

First, let's calculate the total resistance of the circuit:

- R1 = 680 Ω
- R2 = 1.2 kΩ = 1200 Ω
- R3 = 3k3 = 3300 Ω
- R4 = 5.6 kΩ = 5600 Ω
- Voltage source = 12 V

Rtotal = R1 + R2 + R3 + R4

Rtotal = 680 + 1200 + 3300 + 5600

Rtotal = 10780 Ω = 10.78 kΩ

Source Current:

Is = V/Rtotal

Is = 12/2440.5

Is = 0.00492 A = **4.92 mA**

========== **[Theoretical Calculations for KVL (Step-c):]**==========

For R1:

V1 = I × R1

V1 = 0.00111 A × 680 Ω

V1 = 0.7548 V ≈ **0.755 V**

For R2:

V2 = I × R2

V2 = 0.00111 A × 1200 Ω

V2 = 1.332 V ≈ **1.33 V**

For R3:

V1 = I × R3

V1 = 0.00111 A × 3300 Ω

V3 = 3.663 V ≈ 3.66 V

For R4:

V2 = I × R4

V2 = 0.00111 A × 5600 Ω

V4 = 6.216 V ≈ 6.22 V

Verify using Kirchhoff's Voltage Law (KVL)

KVL states that the sum of all voltage drops around a closed loop equals zero.

V1 + V2 + V3 + V4 - Vs = 0 0.755 + 1.33 + 3.66 + 6.22 - 12 = 0 11.965 - 12 = -0.035 ≈ 0

This small difference is due to rounding in our calculations.

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d) Record all the readings mentioned in the previous steps but this time use a DMM to find the measured values.

e) Compare your results and write an equation verifying KVL.

Circuit Entities	V (Calculated)	V (Measured)
R1	0.755V	0.751V
R2	1.33V	1.29V
R3	3.66V	3.54V
R4	6.22V	6.19V
Vs	**12V**	**12V**

To verify KVL, the sum of voltage drops around the closed loop should equal the source voltage:

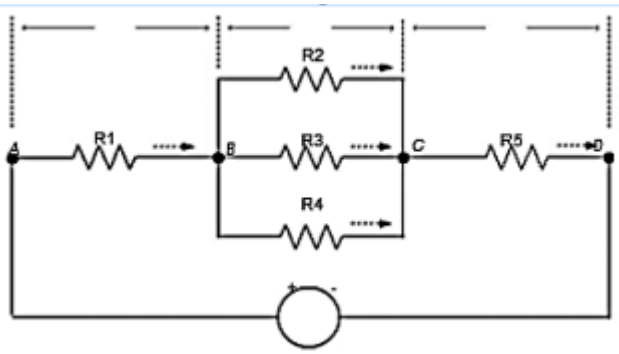
KVL Equation:

V1 + V2 + V3 + V4 = Vs 0.755 V + 1.33 V + 3.66 V + 6.22 V = 11.965 V ≈ **12 V**

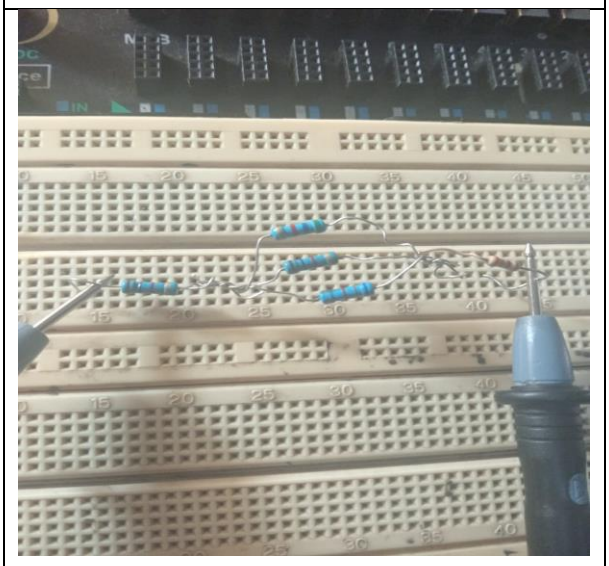
the sum of voltage drops around the closed loop should equal the source voltage:

TASK 2. Solve the following circuit by KCL (at node B):

a) Connect the following circuit on breadboard. Source voltage is 12V and the values of R1, R2, R3, R4 are the same from the previous task. The value of R5 is 1k.



Circuit to prove KCL



b) Before connecting the power supply measure the total resistance of the circuit.

Req(measured) = 2.44K Ω Req(calculated) = 2.26 kΩ

c) Using the measured value of the resistance calculate the source current Is and current through R1, R2, R3, R4 and R5. Record all these values in the calculated section of the following table. Sorce current:4.92mA

========== **[Theoretical Calculations for KVL (Step-c):]**==========

Given values:

- R1 = 6.8 kΩ
- R2 = 1.2 kΩ
- R3 = 3 kΩ
- R4 = 5.6 kΩ
- R5 = 1 kΩ
- Vs = 12V
- **Looking at the parallel section (R2, R3, R4):**
- 1/Rparallel = 1/R2 + 1/R3 + 1/R4
- 1/Rparallel = 1/1200 + 1/3300 + 1/5600
- 1/Rparallel = 0.000833 + 0.000303 + 0.000179
- 1/Rparallel = 0.001315
- 1/Rparallel = 1/0.001315 = 760.5 Ω

TOTAL RESISTANCE:

Rtotal = R1 + Rparallel + R5 Rtotal

Rtotal = 680 + 760.5 + 1000

Rtotal = 2440.5 Ω

Sorce current

Sorce current = V/Rtotal Is = 12/2440.5 Is = 0.00492 A **= 4.92 mA**

Individual Current across erach resistor::

- I1 (through R1) = Is = 4.92 mA (same as source current)
- I5 (through R5) = Is = 4.92 mA (same as source current)

The voltage across the parallel combination: V_parallel = Is × Rparallel V_parallel = 0.00492 × 760.5 V_parallel = 3.74 V

Now calculate currents through the parallel resistors:

- I2 (through R2) = V_parallel/R2 = 3.74/1200 = 0.00312 A = 3.12 mA
- I3 (through R3) = V_parallel/R3 = 3.74/3300 = 0.00113 A = 1.13 mA
- I4 (through R4) = V_parallel/R4 = 3.74/5600 = 0.00067 A = 0.67 mA

Verification of Kirchhoff's Current Law (KCL)

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Table of Results

Circuit Entities	I (Calculated)	I (Measured)
R1	4.92 mA	4.78mA
R2	3.12 mA	3.09mA
R3	1.13 mA	1.10mA
R4	0.67 mA	0.668mA
R5	4.92 mA	4.90mA
Is(Sorce Current)	4.92 mA	4.92mA

KCL equation at node B: I(R1) = I(R2) + I(R4) + I(R5)

At node B: I1 = I2 + I3 + I4 4.92

= 3.12 + 1.13 + 0.67 4.92

= 4.92 ✓

KCL equation:

I1 = I2 + I3 + I4 4.92 mA

= 3.12 mA + 1.13 mA + 0.67 mA

=4.92mA

Conclusions:

After completing laboratory tasks focused on Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL), several key conclusions can be drawn. Firstly, these fundamental laws are absolutely crucial for understanding and analyzing electrical circuits. KCL, which states that the total current entering a junction must equal the total current leaving it, was consistently observed in our experiments. We saw how current divides and recombines at different nodes within a circuit, always maintaining this balance. This reinforced the idea that current is conserved, flowing continuously through the circuit without loss. Similarly, KVL, which dictates that the sum of all voltage drops in a closed loop must equal zero, proved to be equally vital. By meticulously measuring voltage drops across various components and tracing loops within the circuits, we repeatedly verified this principle. We learned that voltage is a measure of potential difference and that it's distributed throughout a loop in a predictable way. Furthermore, the lab tasks highlighted the practical application of these laws in solving real-world circuit problems.